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Practical Fundamentals of Geological Mapping

Introduction to terrain study in Geology

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Welcome to the interactive book **Practical Fundamentals of Geological Mapping**.

This teaching resource has been designed for Earth Sciences students and professionals with the goal of facilitating autonomous and interactive learning of representation and terrain analysis techniques.

Book Structure

The content is organized progressively:

- ***Fundamentals of Mapping***: Key concepts such as strike, dip, dip direction, and the interpretation of different types of geological contacts on maps.
- **Practical Exercises (Coming Soon)**: A battery of solved problems and interactive Python notebooks for the automatic resolution of three-point structural problems and geological cross-sections.

PDF Version (Printable)

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Note

This interactive book is a dynamic project of the **University of Salamanca (USAL)**.

INTRODUCTION TO GEOLOGICAL MAPPING

Learning Objectives

By the end of this chapter, you will be able to:

1. Understand the importance of geological mapping in Earth Sciences.
2. Identify and measure the spatial orientation of geological planes (strike, dip direction, and dip).
3. Interpret basic symbology on a geological map.
4. Solve simple conceptual problems regarding the spatial disposition of terrain.

1.1 1. What is a Geological Map?

A **geological map** is a two-dimensional, scaled representation of the spatial distribution of different rock units (or sediments) outcropping on the Earth's surface, as well as the geological structures that affect them (folds, faults, unconformities).

Unlike a conventional topographic map, a geological map provides a projection of the **third dimension** into the subsurface by interpreting the geometry of contact lines.

1.2 2. Plane Orientation: Strike, Dip, and Dip Direction

To structurally characterize any geological plane (such as a rock layer or a fault), three angular parameters measured with a geologist's compass are used:

- **Strike (Trend of the layer):** The horizontal angle between magnetic north and the line of intersection of the geological plane with a virtual horizontal plane.
- **Dip (Inclination):** The angle of maximum slope measured vertically between the horizontal plane and the geological plane. It ranges between 0° (horizontal) and 90° (vertical).
- **Dip Direction (Sense of inclination):** The direction of the horizontal projection of the line of maximum slope. It is always perpendicular to the strike (90° to the right or left).

1.2.1 Mathematical Relationship of Apparent Dip

If inclination is measured in a direction that is not perpendicular to the strike, an **apparent dip** (α) is obtained, which is always less than or equal to the **true dip** (β). The mathematical relationship is given by:

$$\tan(\alpha) = \tan(\beta) \cdot \sin(\theta)$$

Where:

- α is the apparent dip.
 - β is the true dip.
 - θ is the horizontal angle between the strike direction and the direction in which the apparent dip is measured.
-

1.3 3. Types of Geological Contacts

The geometric relationship between different rock units defines the type of contact. The most common contact relationships are outlined below:

As shown in Fig. 1.1, the diagram is versioned as a static image.

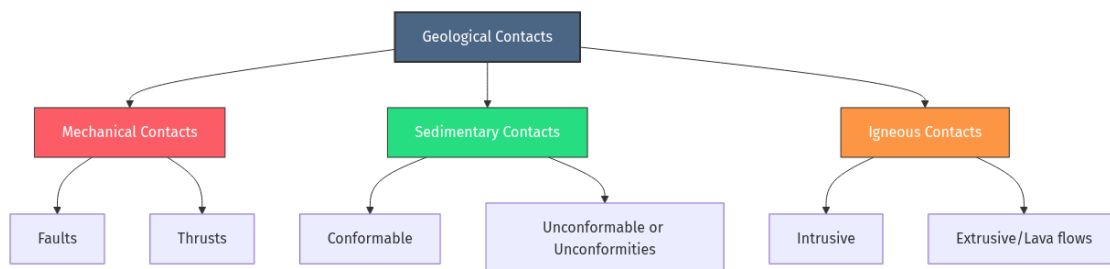


Figure1.1: Diagram generated from code.

1.4 4. The “V” Rule in Mapping

One of the fundamental rules for interpreting the spatial layout of layers from their surface exposure is the **V Rule**:

When an inclined layer crosses a topographic valley, the trace of its contact draws a V shape whose apex points in the direction of the dip of the plane (with minor exceptions where the valley slope is greater than the dip).

- **Horizontal Beds:** Contact traces are parallel to topographic contour lines.
 - **Vertical Beds:** Contact traces are represented as straight lines on the map, completely ignoring the terrain’s topography.
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1.5 5. Self-Assessment Exercises

i Conceptual Question: Apparent Dip

Question: If a layer has a true dip of 30° towards the East, what will be its apparent dip if we observe it in a terrain cut oriented exactly in a North-South direction?

Solution: Since the dip direction is East, the strike of the layer is North-South (perpendicular to East). The direction of observation (North-South) is parallel to the strike of the layer. The angle θ between the strike direction and the direction of observation is 0° .

Applying the formula: $\tan(\alpha) = \tan(30^\circ) \cdot \sin(0^\circ) = 0 \implies \alpha = 0^\circ$

Therefore, on a cross-section plane parallel to the strike, the bed will appear completely horizontal (apparent dip of 0°).

1.6 6. Future Practical Exercises Battery

In upcoming updates to this interactive book, computational tools and executable Python notebooks will be incorporated to solve:

1. **Three-point problem:** Calculation of the strike and dip of a plane from three known elevations.
2. **Geological Cross-Section Generation:** Automated construction of cross-sections from topographic profiles and map traces.
3. **Structure contours:** Automatic tracing of contacts using spatial interpolation.